

Dayflow HRMS

Implementation Decisions & Integration Rules

Internal Engineering Specification

1 Purpose of this Document

This document defines the final technical and architectural decisions required before implementation of the Dayflow Human Resource Management System (HRMS).

It complements the existing SRS and API Contract and does not redefine functional requirements. Its purpose is to remove ambiguity introduced during design and wireframing before coding begins.

2 Scope Compliance with SRS

All implemented features must strictly come from the provided SRS document.

No new functional scope is introduced. Wireframes are treated as a UI representation of SRS requirements, not as a source of new features.

The following core areas remain in scope:

- Authentication and role-based access
- Employee profile management
- Attendance tracking
- Leave and time-off management
- Approval workflows
- Payroll visibility (read-only)

3 User Creation, Signup, and Bootstrap Rules

- Public user signup is **not allowed**
- A single Admin/HR user is seeded by the backend at system initialization
- The wireframe “Sign Up” screen is interpreted as an **Admin-only Add Employee** function
- Employees cannot create accounts for themselves
- All employee accounts are created and managed by Admin/HR users

This resolves the apparent conflict between wireframes and security requirements.

4 Authentication and Credential Rules

- All users authenticate using a system-generated **Login ID**
- Email addresses are stored as profile data and are not used for authentication
- Login IDs are immutable after creation
- Employee passwords are auto-generated by the system at account creation
- Employees may change their password after login
- First-login password enforcement is optional and may be implemented if time permits

5 Login ID Generation Policy

Login IDs are generated automatically by the backend using the following format:

```
OI + first two letters of first name + first two letters of last name  
+ year of joining + serial number for that year
```

This rule originates from the wireframes and is enforced exclusively by backend logic.

6 Role-Based UI and Access Enforcement

- All role-based access rules are enforced by the backend
- Frontend visibility (admin vs employee screens) is a convenience, not a security mechanism
- Any endpoint under `/admin/*` is accessible only to Admin users
- Employees can access only self-related data

Screens explicitly marked as “Admin” in wireframes are treated as Admin-only views.

7 Attendance Status Interpretation

The wireframes introduce visual attendance status indicators.

Attendance status is derived dynamically by the backend using the following rules:

- Approved leave exists for date → `ON_LEAVE`
- Attendance record exists for date → `PRESENT`
- Otherwise → `ABSENT`

Status is not manually editable and should not be inferred by the frontend.

8 Attendance and Payroll Relationship

The wireframes imply that attendance influences payroll calculations.

For the MVP:

- Attendance data is recorded and displayed
- Payroll data is visible in read-only form
- Salary computation based on attendance is **explicitly out of scope**

This decision prevents unintended scope expansion.

9 File Upload Handling

Wireframes depict file uploads (company logo, profile picture).

Implementation decision:

- Profile picture upload may be supported during employee creation if time permits
- Editing or changing uploaded images after creation is out of scope
- Placeholder images may be used if upload functionality is skipped

No file upload functionality is mandatory for MVP completion.

10 Frontend–Backend Responsibility Split

10.1 Frontend Responsibilities

- Render UI based on API responses
- Collect and validate user input
- Invoke backend APIs as defined in the API Contract
- Store and forward authentication tokens

The frontend does not make authorization or business logic decisions.

10.2 Backend Responsibilities

- Authentication and authorization enforcement
- Login ID and password generation
- Business rules and validation
- Workflow and approval decisions
- Database persistence

The backend is the single source of truth.

11 API Communication Rules

- All communication uses JSON over HTTP
- Frontend never accesses the database directly
- All protected endpoints require a valid JWT
- API Contract document is authoritative

12 Error Handling Policy

The backend follows a consistent error-handling approach:

- **200 OK** – Successful request
- **400 Bad Request** – Invalid input
- **401 Unauthorized** – Missing or invalid authentication
- **403 Forbidden** – Insufficient privileges
- **404 Not Found** – Resource does not exist

All errors are returned as JSON.

13 Version Control Policy

- Single shared Git repository
- Single main branch
- Backend and frontend maintained in separate directories
- All contributors commit independently

14 Deferred and Explicitly Out-of-Scope Items

The following items are intentionally deferred or excluded:

- Payroll calculation engine
- Analytics and reports
- Notifications and email workflows
- Advanced search, filtering, and pagination
- Editable salary structures

15 Final Statement

With the SRS, API Contract, and this document in place, all major ambiguities introduced by wireframes have been resolved.

Implementation may proceed without further architectural clarification.